**8.** (i) The curve $C$ has equation $y = g(x)$ where

$$g(x) = e^{3x} \sec 2x \qquad -\frac{\pi}{4} < x < \frac{\pi}{4}$$

(a) Find $g'(x)$

**(2)**

(b) Hence find the $x$ coordinate of the stationary point of $C$.

**(3)**

(ii) A different curve has equation

$$x = \ln(\sin y) \qquad 0 < y < \frac{\pi}{2}$$

Show that

$$\frac{dy}{dx} = \frac{e^x}{f(x)}$$

where $f(x)$ is a function of $e^x$ that should be found.

**(4)**